



Government of Ireland Postgraduate Scholarship Programme 2026 Indicative Applicant Form

This Word document is provided solely for information purposes; please note some differences between this and the final version of the online application form may occur. All applicants must create and complete their submission through the [online system](#) by the relevant deadline. Please see the **2026 Call Document** and the **applicant's Guide to the Online System** on Research Ireland's [website](#) for further information prior to submitting your application online. We recommend that applicants review the "General Application Feedback" available on the aforementioned website. All sections must be completed in full.

PRELIMINARY ELIGIBILITY QUIZ

This is a Preliminary Eligibility Quiz.

The information you input in this quiz will determine your eligibility to apply for the 2026 Government of Ireland Postgraduate Scholarship (GOIPG).

If you do not meet the preliminary Eligibility Criteria, your application will be deemed Ineligible and the system will prevent you from completing a GOIPG Application.

After submitting the Preliminary Eligibility Quiz form, if you have passed this stage, immediately click on 'Continue to Application'. DO NOT navigate out of the page or reload the page before clicking on 'Continue to Application', or before the application form is open. DOING SO WOULD PREVENT YOU FROM OPENING YOUR DRAFT APPLICATION AND FROM RE-SUBMITTING THE PRELIMINARY ELIGIBILITY QUIZ. Research Ireland will not reset the preliminary eligibility quiz.

Once the application is open, it will auto-save after 2 to 3 seconds.

Please note that passing of the preliminary eligibility quiz and provision of access to the application form should not be taken as confirmation that ALL eligibility requirements for the programme have been met. If it becomes apparent to Research Ireland at any stage of the application, assessment or award offer process that an applicant does not meet the eligibility criteria, an application will be deemed ineligible and any award offer shall be withdrawn. Please refer to the GOI Call Document for more information on the Eligibility Criteria.

Question 1: If successful, will you be a new entrant to the postgraduate degree for which you are seeking funding (i.e. due to commence on 1 September 2026), OR have you already commenced the degree with a first date of registration no earlier than 1 September 2025?



Answer: **Yes** or No

If you have answered 'no' to question 1:

If you have a date of first registration prior to 01 September 2025, can you confirm that you did not commence work on your postgraduate degree prior to 1 September 2025, but your institution required you to register before this date. This circumstance must be formally confirmed and endorsed by the relevant Research Body to be considered eligible. Please note that if the relevant Research body cannot confirm and endorse that this is the case, your application will be deemed ineligible, withdrawn from the process and any offer of award shall be rescinded.

OR

If your first date of registration date falls before the designated eligibility window (i.e. before 1 September 2025), do you have a properly documented eligible career break that started before the Call deadline and is at least equivalent to the adjustment being sought? Please note that if the relevant Research body cannot confirm and endorse the eligibility adjustments or if the career break is not at least equivalent to the adjustment being sought, your application will be deemed ineligible, withdrawn from the process and any offer of award shall be rescinded.

Answer: Yes or No

Question 2: Do you have a first class or upper second-class honours bachelor's degree?

Answer: Yes or **No** or Bachelor's degree results unknown at time of application or Master's degree results unknown at time of application

If you have answered 'no' to question 2:

Do you have a master's degree?

Answer: **Yes** or No

Question 3: Please indicate the type of scholarship you are applying for:

Answer: Master's by Research Degree or **Doctoral Degree**.

If 'Master's by Research Degree' is selected: Have you previously been awarded a Research Master's, Structured Research Master's, or Taught Master's degree, regardless of the research topic or funding source (including but not limited to Research Ireland)?



Answer: Yes or No

If 'Doctoral Degree' is selected: Have you previously been awarded a doctoral degree (PhD), regardless of the research topic or funding source (including but not limited to Research Ireland)?

Answer: Yes or **No**

Question 4 : Have you had two previous unsuccessful applications to the Government of Ireland Postgraduate Scholarship Programme (including strategic partner themes)?

Answer: Yes or **No**

Question 5: Do you hold, or have you previously held, an Research Ireland scholarship?

Answer: Yes or **No**

If answered 'yes' to question 5: Was the scholarship held to complete a master's or a PhD degree?

Answer: Master's degree or PhD degree

If 'master's degree' is selected: Is this application to complete a PhD degree?

Answer: Yes or No

Question 6: Does your proposed research deal with any of the following prohibited areas?

- Research activity aimed at human cloning for reproductive purposes
- Research activity intended to modify the genetics of human beings that could make such changes heritable (with the exception of research relating to cancer treatment of the gonads, which may be funded)
- Research activities to create human embryos solely for the purpose of research or for the purpose of stem cell procurement, including by means of somatic cell nuclear transfer

Answer: Yes or **No**

Question 7: Has a primary supervisor, based in an eligible Irish Research Body or eligible Irish Research Performing Organisation, agreed to act as supervisor for the above project?



PLEASE NOTE: All applications require a primary supervisor who is willing to supervise the proposed research project. Applicants **MUST** contact their primary supervisor and research office before completing a preliminary eligibility quiz or submitting an application form. Please do **NOT** add a primary supervisor to your application without their prior approval. **RESEARCH IRELAND IS NOT IN A POSITION TO ASSIST APPLICANTS IN FINDING AN ACADEMIC SUPERVISOR FOR THEIR APPLICATION.**

PLEASE NOTE: Primary supervisors may not support more than one applicant to the Government of Ireland Postgraduate Scholarship¹. As such, Primary Academic Supervisors should endorse only one submitted application. If the Research endorses multiple applications with the same primary supervisor, all applications will be deemed ineligible.

Answer: **Yes** or No

If answered 'yes' to question 7: Please provide the name of your primary supervisor (to be selected from a list of primary supervisors registered on the online system by the Irish Research Body)

PLEASE NOTE: Please note that your application will be deemed **INELIGIBLE** if no primary supervisor name is included in the preliminary eligibility quiz, the name of the primary supervisor included in the preliminary eligibility quiz on the online system does not match that on the application form, or if you list more than one primary supervisor. **Make sure to confirm with your primary supervisor the name (and email account) they are registered within the online system.**

*If answered 'no' to question 7: All applications require a primary supervisor who is willing to supervise the proposed research project. Prior to creating an application, you **must** contact and discuss your research project with your proposed supervisor(s).*

APPLICATION FORM

In addition to the Government of Ireland Postgraduate Scholarship, a number of themed scholarships are offered in collaboration with our strategic funding partners as a mechanism to deliver on our shared national objectives. While applicants may apply for one scholarship theme only, those who choose to apply for a strategic funding partner scholarship will also be considered for a Government of Ireland Postgraduate Scholarship. Full descriptions of each of the strategic funding partner scholarships can be found in the Call Document available on the [GOIPG website](#).

¹ This condition relates only to applications to the current open Call and not to any existing scholar the Supervisor may be supporting under previous Government of Ireland Postgraduate Scholarship Calls.



Application details

Project Title: [Semi-Classical and Microlocal Analysis of Szegő Kernels and Toeplitz Quantizations](#).

Irish Research Body: [Trinity College Dublin](#)

Application in Irish: 'yes' or '[no](#)'

There is an option on the system at this point to upload an English translation should you wish to do so. This will accompany your research proposal. The system will only accept documents in PDF format.

Applicant details

ORCID identifier: [0009-0004-5051-3899](#)

ORCID ID provides a persistent digital identifier that distinguishes you from every other researcher. If you do not currently have an ORCID ID, please register for one at www.orcid.org and provide us with your unique 16-digit identifier.

What is your nationality, i.e. your passport-issuing country: [Taiwan](#)

If you are from the Republic of Ireland or Northern Ireland, which county are you from?

Are you a national of a European Union member state (including the Republic of Ireland) OR Iceland, Norway, Liechtenstein, Switzerland, the United Kingdom or Ukraine? 'yes' or '[no](#)'

Will you have been ordinarily resident* in a European Union member state (including the Republic of Ireland) OR Iceland, Norway, Liechtenstein, Switzerland, the United Kingdom or Ukraine for a continuous period of three of the five years preceding 1 October 2025? 'yes' or '[no](#)'

**Ordinarily resident refers to the applicant's place of legal and permanent residence. This will not be deemed as having been interrupted if an absence from the country of ordinary residency has been caused by the training, education or employment of the applicant or the applicant's spouse or parents.*

- If both answers 'yes': YOU WILL BE ASSESSED AS A CATEGORY 1 APPLICANT (for more information see Call Document, section 6.2).
- If one or both answers 'no': YOU WILL BE ASSESSED AS A CATEGORY 2 APPLICANT (for more information see Call Document, section 6.2).

PRIMARY AND ADDITIONAL PARTICIPANTS



*The Irish Research Body and Primary Supervisor details selected in the preliminary eligibility quiz must match those entered into the application form. Applicants should ensure they have contacted the Primary Supervisor and that the Primary Supervisor is willing to support the application before submitting the preliminary eligibility quiz or the application. **Applications that are not endorsed by the academic Primary Supervisor will be deemed ineligible and will not progress. Research Ireland is not responsible for connecting applicants with potential Primary Supervisors and WILL NOT assist in this process.***

Applicants are reminded that Primary Supervisor may support no more than one applicant to the Government of Ireland Postgraduate Scholarship programme. As such, Primary Supervisor should endorse only one submitted application. If the Research endorses multiple applications with the same Primary Supervisor, all applications will be deemed ineligible.

Primary supervisor details – to be selected from a list of primary supervisors registered on the online system by the Irish Research Body:

Name: [Nikhil Savale](#)

Irish research body: [School of Mathematics, Trinity College Dublin](#)

Email address: SAVALEN@tcd.ie

Secondary supervisor details (if applicable):

Name:

Higer Education Institution:

Email address:

TRACK RECORD

Please refer to the [National Framework of Qualifications](#) for the classification of all degrees.

Bachelor's degree (or equivalent):

If your degree results refer to a grading system other than the Irish honours system (first class honours, upper second class (2.1) honours etc.) or the 4.0 grade point average scale, please explain the grading system being referred to in the free text box provided.

Institution: [National Taiwan University](#)

Graduation date: [30/06/2023](#)



Qualification type and name: **Bachelor of Science**

Is this degree completed? **'yes'** or **'no'**

If answered 'yes': Final grade or grade point average: **3.45**

If answered 'no': Please include the grade from the last completed year of the degree in question:

Please note that any conditional offer will be dependent on the final grade for this degree meeting the eligibility requirement of the programme.

All degree results as entered in the application form and endorsed by the Irish Research Body must be the applicant's overall results and verifiable as such on official transcripts. Regardless of whether they are relevant to the eligibility requirements for the programme, conditional awardees will be required to submit official transcripts for each degree listed in the application confirming the overall result, date of graduation and awarding institution. If such transcripts cannot be presented, or do not verify the information exactly as detailed in the application, any conditional offer of an award will be withdrawn. Supporting, non-returnable material must be forwarded to Research Ireland for inspection before the award can commence. All transcripts must be appropriately stamped and signed to confirm that they are official documents of the institution.

Any additional information relating to this degree or your final grade can be included here: **[Max 300 words]**

According to NARIC Ireland Foreign Qualifications, a Taiwanese bachelor's degree is equivalent to an Honours Bachelor's Degree at NFQ Level 8.

A GPA of 3.45 corresponds to an 80.88 percentage grade.

The lowest passing grade for undergraduate students is C-(60%).

The following is the Letter Grade–Grade Point–Percentage Grade Equivalent and the definitions of letter grades.

A+: 4.3, 90-100%, All goals achieved beyond expectation

A: 4.0, 85-89%, All goals achieved

A-: 3.7, 80-84%, All goals achieved, but need some polish

B+: 3.3, 77-79%, Some goals well achieved

B: 3.0, 73-76%, Some goals adequately achieved

B-: 2.7, 70-72%, Some goals achieved with minor flaws

C+: 2.3, 67-69%, Minimum goals achieved

C: 2.0, 63-66%, Minimum goals achieved with minor flaws

C-: 1.7, 60-62%, Minimum goals achieved with major flaws

F: 0, 59% or below, Minimum goals not achieved

X: 0, 0%, Not graded due to unexcused absences or other reasons

**Second bachelor's degree (if applicable):**

If your degree results refer to a grading system other than the Irish honours system (first class honours, upper second class (2.1) honours etc.) or the 4.0 grade point average scale, please explain the grading system being referred to in the free text box provided.

Institution:

Graduation date:

Qualification type and name:

Is this degree completed? 'yes' or 'no'

If answered 'yes': Final grade or grade point average:

If answered 'no': Please include the grade from the last completed year of the degree in question:

Please note that any conditional offer will be dependent on the final grade for this degree meeting the eligibility requirement of the programme.

All degree results as entered in the application form and endorsed by the Irish Research Body must be the applicant's overall results and verifiable as such on official transcripts. Regardless of whether they are relevant to the eligibility requirements for the programme, conditional awardees will be required to submit official transcripts for each degree listed in the application confirming the overall result, date of graduation and awarding institution. If such transcripts cannot be presented, or do not verify the information exactly as detailed in the application, any conditional offer of an award will be withdrawn. Supporting, non-returnable material must be forwarded to Research Ireland for inspection before the award can commence. All transcripts must be appropriately stamped and signed to confirm that they are official documents of the institution.

Any additional information relating to this degree or your final grade can be included here: **[Max 300 words]**

Master's degree (if applicable):

If your degree results refer to a grading system other than the Irish honours system (first class honours, upper second class (2.1) honours etc.) or the 4.0 grade point average scale, please explain the grading system being referred to in the free text box provided.

Type: 'Taught' or 'Research'

Institution: [National Taiwan University](#)

Graduation date: [30/06/2025](#)



Qualification type and name: **Master of Science**

Is this degree completed? **'yes'** or **'no'**

If answered 'yes': Final grade or grade point average: **3.81**

If this degree is unfinished at the time of the application, evidence of successful completion will be required through the submission of transcripts.

All degree results as entered in the application form and endorsed by the Irish Research Body must be the applicant's overall results and verifiable as such on official transcripts. Regardless of whether they are relevant to the eligibility requirements for the programme, conditional awardees will be required to submit official transcripts for each degree listed in the application confirming the overall result, date of graduation and awarding institution. If such transcripts cannot be presented, or do not verify the information exactly as detailed in the application, any conditional offer of an award will be withdrawn. Supporting, non-returnable material must be forwarded to Research Ireland for inspection before the award can commence. All transcripts must be appropriately stamped and signed to confirm that they are official documents of the institution.

Any additional information relating to this degree or your final grade can be included here: **[Max 300 words]**

According to NARIC Ireland Foreign Qualifications, a Taiwanese master's degree is equivalent to a master's degree at NFQ Level 9.

This is a Master's degree with a substantial research thesis.

A GPA of 3.81 corresponds to an 85.83 percentage grade.

The lowest passing grade for graduate students is B-(70%).

The following is the Letter Grade–Grade Point–Percentage Grade Equivalent and the definitions of letter grades.

A+: 4.3, 90-100%, All goals achieved beyond expectation

A: 4.0, 85-89%, All goals achieved

A-: 3.7, 80-84%, All goals achieved, but need some polish

B+: 3.3, 77-79%, Some goals well achieved

B: 3.0, 73-76%, Some goals adequately achieved

B-: 2.7, 70-72%, Some goals achieved with minor flaws

C+: 2.3, 67-69%, Minimum goals achieved

C: 2.0, 63-66%, Minimum goals achieved with minor flaws

C-: 1.7, 60-62%, Minimum goals achieved with major flaws

F: 0, 59% or below, Minimum goals not achieved

X: 0, 0%, Not graded due to unexcused absences or other reasons

**Second master's degree (if applicable):**

If your degree results refer to a grading system other than the Irish honours system (first class honours, upper second class (2.1) honours etc.) or the 4.0 grade point average scale, please explain the grading system being referred to in the free text box provided.

Type: 'Taught' or 'Research'

Institution:

Graduation date: dd/mm/yyyy

Qualification type and name:

Is this degree completed? 'yes' or 'no'

If answered 'yes': Final grade or grade point average:

If this degree is unfinished at the time of the application, evidence of successful completion will be required through the submission of transcripts.

All degree results as entered in the application form and endorsed by the Irish Research Body must be the applicant's overall results and verifiable as such on official transcripts. Regardless of whether they are relevant to the eligibility requirements for the programme, conditional awardees will be required to submit official transcripts for each degree listed in the application confirming the overall result, date of graduation and awarding institution. If such transcripts cannot be presented, or do not verify the information exactly as detailed in the application, any conditional offer of an award will be withdrawn. Supporting, non-returnable material must be forwarded to Research Ireland for inspection before the award can commence. All transcripts must be appropriately stamped and signed to confirm that they are official documents of the institution.

Any additional information relating to this degree or your final grade can be included here: **[Max 300 words]**

Other education:

Please provide any additional information relevant to your academic background which should include the name, location and date(s) of any training courses attended: **[Max 300 words]**

During my undergraduate studies at National Taiwan University, I also participated in the Undergraduate Summer Research Program (USRP) and Undergraduate Research Program (URP) hosted by NCTS (National Center for Theoretical Sciences) from July 2022 to June 2023. Under the supervision of Professor Hsiao, I worked on projects concerning heat and Bergman kernels in complex geometry, gaining experience in reading research papers, developing mathematical arguments, and presenting results. These early experiences were



crucial in shaping my ability to conduct independent research and pursue further study in complex geometric analysis.

More recently, during my graduate studies, I attended the Special Geometric Structures and Analysis summer school held at St. Mary's College, California, in June 2024, organised by SLMath. The program offered intensive lectures on symbol spaces, pseudodifferential operators, and geometric measure theory. This training deepened my understanding of analytic techniques that later became important to my graduate research.

Subsequently, I had the opportunity to visit Professor George Marinescu at the University of Cologne, supported by NCTS, from August to November 2024. During this visit, I learned from Professors Marinescu and Savale, which provided valuable insight into ongoing developments in complex geometry and spectral analysis.

Research achievements:

Please provide any additional information regarding your research achievements to date such as publications, research awards, creation of data sets and databases, conference papers, patents, excavations, public broadcasts, stage performances, creative writing, creative productions and/or exhibitions: **[Max 300 words]**

My master's thesis, which received the Dean's Award from the College of Science at National Taiwan University, focused on the analytic structures underlying the Bergman kernel. Under the supervision of Professor Hsiao, I explored complex geometry and microlocal analysis through this lens. My work centers on three main directions, each of which I plan to develop further.

First, we looked at the Bergman kernel's leading behavior, which usually needs positivity and heavy methods. We introduced a flexible "local spectral gap" condition, using scaled coordinates and differential equations to find the main asymptotic behavior. As the parameter k increases, the scaled kernel approaches a quadratic weight. This method, when extended to complex orbifolds, simplifies Tian's 1990 results and suggests uses for irregular metrics, guiding my ongoing research.

Second, we establish a simpler way to find the full asymptotic expansion of the Bergman kernel, revealing detailed geometric information. Using the Melin-Sjöstrand stationary phase formula, we compared the kernel to a quadratic model, recovering the expansion and calculating the first two coefficients, which match known results. The leading coefficient involves the determinant of the curvature operator, and the next includes scalar curvatures. This framework may be adaptable to singular or non-compact settings, which I intend to explore further.

Third, we advanced Toeplitz quantization, linking complex geometry to quantization. Beyond functions, we applied it to pseudodifferential operators, including momentum data important for microlocal analysis and physics. Using the stationary phase, we derived a full asymptotic



expansion and composition formula, broadening classical results. This opens paths to new quantization frameworks.

Work experience:

Please provide details of any relevant work experience, including voluntary work, to date which should include employers' names, job titles, nature of duties and responsibilities, as well as duration of employment: **[Max 300 words]**

During the fall semester of my final undergraduate year, I served as a Teaching Assistant for Complex Analysis (Honors Course). My responsibilities included grading assignments and holding weekly office hours to assist students with problem-solving and conceptual understanding. In the fall semester of my first year in the master's program, I was a Teaching Assistant for Introduction to Geometry, where I additionally conducted a biweekly tutorial session to review key concepts and guide students through exercises.

Between the completion of my master's degree and the commencement of my national military service, I have been serving as a Research Assistant at the National Center for Theoretical Sciences (NCTS), Mathematics Division. My work continues the direction of my master's research in complex geometry and microlocal analysis, focusing on developing analytic tools related to the Bergman kernel and geometric quantization. This position allows me to dedicate myself full-time to research. I also attend seminars and reading groups that broaden my perspective on geometric analysis and related areas. This experience gives me a chance to work as a researcher. It has strengthened my capacity for self-directed inquiry, sharpened my technical and writing skills, and confirmed my long-term commitment to pursuing an academic career in pure mathematics.

PERSONAL STATEMENT

Please highlight any additional information which has not been included elsewhere in the application, e.g.:

- Why do you wish to pursue a higher degree by research?
- Why have you proposed this research topic?
- Why do you feel there is a specific demand for the skill set that you wish to build?
- Why are you particularly suited to this research field?
- Which of your attributes demonstrate your capability to be a good researcher, e.g. motivation, commitment, thirst for knowledge?

[Max 500 words]

My interest in mathematics has always been driven by the balance between beauty and structure. While the Riemann Hypothesis first drew me into mathematics, it was the distinctive nature of complex geometry that made me want to pursue research. Unlike Riemannian or differential geometry, complex geometry is shaped by holomorphicity and symmetry, leading to properties that are both unexpected and elegant. At the same time, I was



intrigued by complex dynamical systems, particularly by fractal structures, which revealed how even simple rules can generate profound complexity. Together, these experiences convinced me that I wanted to explore the theories where symmetry, geometry, and analysis interact.

This naturally led me to focus on kernel theory and microlocal analysis. Kernel theory is compelling because it builds a bridge between geometry and analysis. It transforms geometric questions about functions into analytic ones and allows geometric quantities to be read from analytic expansions. Microlocal analysis provides a flexible and powerful framework to extract information from cotangent bundles and to work with differential operators in a broader sense. The combination of these perspectives not only enables a more complete analysis of geometric objects but also creates opportunities to connect seemingly different mathematical structures. I believe this synthesis has continuing demand within mathematics, as it offers tools that are both conceptually deep and broadly applicable.

I consider myself well-suited to contribute to this field. I work with strong intuition and a willingness to take bold steps. For example, when developing a method for coefficient computations, I extended the domain of integration beyond what seemed manageable and discovered a surprisingly simple result. At the same time, I am committed to expanding my knowledge, such as studying Donaldson-Sun theory, even when it requires me to venture beyond familiar ground. My persistence, curiosity, and readiness to engage with difficult ideas have shaped me into a researcher who is motivated not only by technical challenges but also by the pursuit of new perspectives. I see research as both an exploration of mathematical beauty and a way to build bridges across theories, and I am eager to continue this journey at the doctoral level.

CAREER TRAINING AND DEVELOPMENT PLAN

Please provide a career training and development plan which addresses the following:

- What are your career goals and how would this scholarship help you to achieve them?
- How will you go about acquiring the expert knowledge and transferable skills necessary for your professional development, e.g. technical skills, communication skills, analytical skills?
- How would this scholarship enable you to gain skills relevant to employment outside the traditional academic sector?
- How can the scholarship transform your existing skills in those identified as being required to pursue the chosen career?

[Max 1000 words]

My long-term career goal is to become an independent researcher who contributes innovative and impactful results in the interconnected field of complex geometry and analysis. Simultaneously, I aim to maintain a solid and active international network. Achieving this goal necessitates not only deepening my expertise in advanced mathematical theories but also



cultivating a diverse set of transferable skills that are essential for both academic and non-academic career paths. This scholarship presents a pivotal opportunity: it would enable me to dedicate myself fully to research, learn from esteemed scholars in the field, and build the professional profile necessary for a sustainable academic career.

On the professional front, my development plan focuses on expanding my mathematical foundation across several interconnected domains. Building upon my current research in complex geometry, I intend to extend my knowledge to Cauchy-Riemann (CR) geometry, contact geometry, and conformal geometry. These frameworks serve as crucial tools for studying the interface between geometric and analytic structures. Mastering these theories will equip me with the conceptual flexibility to address problems that link local analytic behavior with global geometric properties. Additionally, I plan to enhance my understanding of invariants and local index theorems, which bridge analysis and topology and are essential for comprehending quantization phenomena. By exploring how curvature and spectral data encode geometric information, I hope to discover novel connections between microlocal and global perspectives.

An equally important direction is my training in microlocal analysis. I plan to move beyond classical pseudodifferential operators to develop a comprehensive understanding of Fourier integral operators and their symbolic structures. This will enhance my ability to study the fine geometry of kernels and to analyze oscillatory phenomena that appear in quantization problems. Through this training, I intend to acquire not just technical mastery but also a sharper intuition for how local analytic structures reflect underlying geometric symmetries. Ultimately, these skills will allow me to design new analytic frameworks adaptable to complex or singular settings, which are essential for making original and independent contributions in geometric analysis.

To achieve these goals, I will pursue advanced courses, attend specialized workshops, and seek guidance from mentors whose expertise complements my own. Regular interaction with scholars working in global analysis and mathematical physics will expose me to diverse methodologies, helping me learn how to translate geometric ideas into analytic tools and vice versa. This cross-training will develop both the depth and versatility required of a modern researcher in mathematics: the ability to think abstractly, reason analytically, and communicate precisely.

In parallel with technical training, I will focus on strengthening key transferable skills that are increasingly vital for success across all professional sectors. Effective communication is my foremost priority. I aim to refine my academic writing style to produce papers that are not only rigorous but also conceptually clear and persuasive, inviting engagement from readers across related fields. I will also practice delivering presentations that combine clarity with narrative coherence, enabling me to explain complex ideas to audiences with varied backgrounds. These communication skills are equally valuable in research, interdisciplinary collaboration, and industry, where clarity of thought and persuasion are fundamental.

Another dimension of my plan is to cultivate international collaboration and project management skills. Research today depends on the ability to organize complex, long-term



projects while maintaining creative momentum and adaptability. By engaging in collaborative research, coordinating seminar activities, and participating in international programs, I will learn how to balance individual initiative with collective progress. These experiences will strengthen my ability to plan strategically, allocate effort effectively, and work productively in diverse teams, which is directly transferable to leadership roles within or beyond academia.

This scholarship will be transformative in uniting these elements. Its support will give me access to top-tier mentorship, advanced research facilities, and structured training programs designed to develop both technical expertise and transferable competencies. The financial stability provided by the scholarship will allow me to focus entirely on research, take intellectual risks, and invest in long-term skill development.

Ultimately, this scholarship will enable me to transform my existing strengths, for instance, intuition, persistence, and boldness, in exploration into a coherent professional skill set grounded in analytical precision, clear communication, and effective collaboration. It will equip me to pursue a lifelong career of discovery, one that bridges theory and application, and contributes both to the advancement of mathematics and to the broader community that values critical and creative thought.

PROPOSED RESEARCH

If successful, will you be a new entrant to the degree for which you are seeking Research Ireland funding on 1 September 2026? **'yes'** or **'no'**

If answered 'yes': Please specify the scholarship type for which you are applying:

*'24-month Master's by Research Degree' or **'48-month Doctoral Degree'**.*

If answered 'no': Please specify the degree type which you intend to complete: *'24-month Master's by Research Degree' or '48-month Doctoral Degree'.*

If answered 'no': What date did you first register for this degree?

It is important that the information in this field is entered correctly as the duration of funding awarded is dependent on the date of first registration.

For applicants who have already completed part of the degree for which they are seeking Research Ireland funding, the duration of funding to be awarded is dependent on the type of postgraduate degree being pursued and the date of first registration as detailed in the call document.

If answered 'no': Please note if the date of first registration that you have provided falls outside the eligible window indicated in the call document for the degree type you are pursuing, you must provide details of the eligible career break in the box below so that an adjustment to your eligibility may be considered by Research Ireland.



Do you require an eligibility adjustment on the grounds of a career break?
'yes' or 'no'

If answered 'yes': Applicants whose date of first registration falls before the eligibility window for the degree which they are pursuing may still be deemed eligible for the Call if they have a properly documented eligible career break, provided the eligible career break started before the call deadline and is at least equivalent to the adjustment being sought. Eligible career breaks include maternity leave, paternity leave, adoptive leave, parental leave, parent's leave, extended sick leave and carer's leave.

If an applicant wishes to adjust their period of eligibility, the Research Body must confirm that the individual meets the requirements for the adjusted period of eligibility as per the time of the eligible break(s). Documented evidence of eligible career break should not be sent to Research Ireland unless requested. **Please do not provide any personal health and/or medical documentation and/or information (including evidence of a medical diagnosis/diagnoses) in correspondence with Research Ireland.**

Please provide details of eligible career break below:

Maternity Leave: [dd/mm/yyyy- dd/mm/yyyy](#)

Paternity Leave: [dd/mm/yyyy- dd/mm/yyyy](#)

Adoptive Leave: [dd/mm/yyyy- dd/mm/yyyy](#)

Parental Leave: [dd/mm/yyyy- dd/mm/yyyy](#)

Prolonged Sick Leave: [dd/mm/yyyy- dd/mm/yyyy](#)

Please indicate the reason(s) for requesting this extension, the number of months sought, and the date(s) when the career break(s) took place.

If answered 'no': Please indicate the progress of your research to date, i.e. what has been done and what still needs to be done. Please also indicate whether or not you have been confirmed on the PhD register at your host institution: **[Max 500 words]**

If answered 'no': Please provide details of your current funding source: **[Max 300 words]**

Will you be in receipt of any additional awards or research / academic salary during the scholarship period, e.g. scholarships, positions, bursaries, travel grants etc.? *'yes' or 'no'*

Failure to provide this information may render your application ineligible.

If answered 'yes': Please provide further details, including the value: **[Max 300 words]**

Failure to provide this information may render your application ineligible.



Have you previously submitted a proposal to the Government of Ireland Postgraduate Scholarship scheme and been unsuccessful? 'yes' or '**no**'

If answered 'yes': Did it contain all or part of the present proposal? 'yes' or '**no**'

If answered 'yes': Please clearly describe the modifications to your research proposal since it was previously submitted:

[Max 300 words]

Do you currently hold, or have you previously held, any other Research Ireland awards? 'yes' or '**no**'

If answered 'yes': Please provide further details **[Max 300 words]**

Primary area: [Mathematics](#)

Please consult the [research categorisation document](#) available on Research Ireland website for further descriptions of primary area, discipline and other research area

Discipline: [Pure Mathematics](#)

Please consult the [research categorisation document](#) available on Research Ireland website for further descriptions of primary area, discipline and other research area

Other research area(s):

- [Algebraic and Differential Geometry](#)
- [Partial Differential Equations](#)
- [Real and Complex Functions](#)

Please consult the [research categorisation document](#) available on Research Ireland website for further descriptions of primary area, discipline and other research area

Second categorisation – if interdisciplinary:

Please consult the [research categorisation document](#) available on Research Ireland website for further descriptions of primary area, discipline and other research area

Keywords describing proposed research:

- [Bergman and Szegő kernels](#)
- [Cauchy-Riemann Geometry](#)
- [Fourier Integral Operator](#)



- Geometric Quantization
- Pseudodifferential Operator
- Toeplitz Operator

Please provide a lay abstract for your proposed research, which will be used to inform a non-specialist audience:

[Max 300 words]

Mathematics often endeavors to describe the shape and behavior of spaces, ranging from simple surfaces to highly curved, multidimensional realms. My research delves into such spaces when they are constructed from complex numbers, known as complex manifolds. These spaces hold significant importance in modern geometry and theoretical physics as they encapsulate both symmetry and curvature in a unified language.

A crucial tool in comprehending these spaces is the reproducing kernel, a mathematical function that reflects how information propagates and interacts within the space. It can be likened to a fingerprint that encodes the local geometry and enables the reconstruction of global properties. By meticulously analyzing the intricate structure of these kernels, we gain profound insights into the behavior of geometric spaces at both small and large scales.

The concept of quantization serves as a bridge between the geometric and analytic perspectives. In this context, quantization entails translating geometric data, such as curvature or symplectic form, into algebraic operators that operate on spaces of functions or sections. Grasping this correspondence necessitates precise control over the reproducing kernel and its microlocal structure.

My project aims to develop precise formulas and analytic tools for these kernels, thereby extending quantization theory to broader geometric settings. This endeavor not only deepens our comprehension of the concealed structures within complex spaces but also contributes to the mathematical foundations of quantum mechanics and geometric analysis.

Please provide details of your proposed research to include (a) aims, objectives and central research questions of the project, (b) how existing literature on the topic has been used to inform the proposal and (c) how the project will advance state of the art and make a contribution to existing knowledge:

[Max 500 words]

This proposed research centers on advancing the analytic and geometric understanding of the Szegő kernel within the framework of Cauchy-Riemann (CR) geometry. The Szegő kernel, as a reproducing kernel for CR functions, plays an important role in several complex variables and geometric quantization. Its asymptotic expansions encode deep geometric and spectral information, including curvature invariants and boundary phenomena, yet many aspects, especially explicit formulas and higher-order coefficients, remain only partially understood, particularly in non-integrable or higher codimension CR manifolds. The primary aim of this project is to derive explicit asymptotic expansions for the Szegő kernel and associated



Toeplitz operators on CR manifolds, beginning with model cases such as the Heisenberg group. This involves constructing parametrices for the Kohn Laplacian, analyzing their semiclassical behavior as the tensor power increases, and developing a microlocal and symbolic calculus tailored to CR settings. Central research questions include: How do the geometric and spectral properties of CR manifolds manifest in the asymptotic behavior of the Szegő kernel? Can explicit formulas and recurrence relations for Szegő kernel coefficients be established, and what geometric invariants do they encode? How can microlocal analysis and semiclassical techniques be unified to study quantization phenomena in CR geometry?

This proposal builds upon a rich and well-established foundation of prior research. Hörmander and Boutet de Monvel laid the analytic groundwork by proving hypoellipticity and constructing parametrices for the Kohn Laplacian. Subsequently, Boutet-Sjöstrand and Hsiao advanced the field by introducing a Fourier integral operator framework that unveiled the Szegő kernel's intricate microlocal structure. While these contributions offer profound structural insights, they remain predominantly qualitative: explicit formulas and concrete higher-order expansions are still elusive, particularly in broader CR geometric contexts. Bridging this gap requires refined analytical techniques capable of producing computable expansions and elucidating their connections to underlying geometric invariants. In this vein, my previous work on the Bergman kernel and Toeplitz quantization opens promising directions, as analogous methods may be adapted to derive explicit representations and uncover deeper geometric interpretations of the Szegő kernel.

The project will advance the state of the art by providing explicit asymptotic expansions and formulas for the Szegő kernel and Toeplitz operators on CR manifolds, addressing gaps in the literature where only partial results exist. By developing a refined microlocal and symbolic calculus for CR geometry, the research will enable systematic computation of kernel coefficients and their geometric interpretation, potentially revealing new CR invariants. Integrating semiclassical analysis and Fourier integral operator techniques will unify approaches to quantization in CR settings, bridging analytic and geometric perspectives. These results will lay the groundwork for further applications in contact and conformal geometry, and for exploring connections with mathematical physics, such as quantization and spectral theory on non-standard geometric backgrounds. Overall, this research will deepen the understanding of the Szegő kernel as a central object in CR geometry, advance analytic techniques for kernel expansions, and contribute new tools and invariants to the field.

Please detail the research design and methodologies to be employed in carrying out your scholarship which should be described in sufficient detail to demonstrate your thorough understanding of the research topic:

[Max 500 words]

To address the core problems in kernel asymptotics and quantization across CR geometry, I will employ a combination of analytic and microlocal techniques. My research design centers on three main methodologies:



I will develop and apply the CR scaling method to study the asymptotic behavior of Szegő kernels and Toeplitz operators on CR manifolds and high tensor powers CR line bundle. By rescaling coordinates and weights near a point, this method allows the reduction of complex geometric problems to model cases, such as the Heisenberg group. The semi-classical approximation will be used to analyze the behavior of kernels and operators as the quantization parameter, which is usually the tensor power of the CR line bundle, tends to infinity, revealing leading-order terms and geometric invariants. This approach provides explicit formulas and enables the computation of full expansions by systematically tracking the dependence on the scaling parameter.

Building on the foundational work of Boutet de Monvel, I will construct parametrices for the Kohn Laplacian and related operators on CR and contact manifolds. This involves developing symbolic calculus for subelliptic operators and constructing explicit Fourier integral operator representations of the Szegő kernel. The parametrix construction is crucial for understanding hypoellipticity, spectral properties, and for deriving asymptotic expansions of kernel functions. I will adapt these techniques to more general geometric settings, including non-integrable CR structures and sub-Riemannian Laplacians, to obtain new analytic tools and invariants.

The stationary phase method is a powerful tool for extracting asymptotic expansions from oscillatory integrals, which naturally arise in the analysis of kernel functions and Toeplitz operators. I will use the Melin-Sjöstrand stationary phase formula to compare kernels with quadratic models and to compute explicit coefficients in their expansions. This method allows the identification of geometric quantities encoded in the leading and higher-order terms, such as curvatures and bracket structures. By refining the stationary phase analysis to accommodate singularities and boundary effects, I aim to extend its applicability to non-compact and singular settings.

These methodologies will be integrated to provide a unified analytic framework for kernel asymptotics and quantization. The CR scaling and semi-classical approximation will guide the reduction to model cases and the identification of leading behavior. Boutet de Monvel's parametrix construction will supply explicit operator representations and symbolic expansions. The stationary phase method will enable the computation of full asymptotic series and the extraction of geometric invariants. Together, these approaches will allow me to address open problems in kernel expansions, quantization, and the construction of new invariants in complex, CR, contact, and conformal geometries.

Please provide a schedule to include (a) milestones and deliverables for completion of the proposed research, (b) risks that might endanger reaching these deliverables and (c) the contingency plans to be put in place in order to mitigate these risks:

[Max 500 words]

The proposed research will proceed in several stages, beginning with the model case and advancing toward increasingly general and challenging settings. In the initial phase, I will



focus on explicit analysis of Szegő and Toeplitz kernels in the model CR case, such as the Heisenberg group, where semiclassical and microlocal techniques can be fully developed and tested. Building on these foundations, the next milestone is to extend these methods to general CR manifolds, refining stationary phase and symbolic calculus to capture the full asymptotic expansions and geometric invariants. If progress and circumstances allow, I aim to further generalize these results to contact and conformal geometries, constructing Szegő-type and Toeplitz-type operators and investigating their invariants and spectral properties.

Several risks may arise during this process. The technical complexity of moving from the model case to general CR, contact, or conformal settings may present unforeseen analytical obstacles, especially in non-integrable or singular situations. Explicit computation of higher-order coefficients and symbolic expansions may prove more intricate than anticipated, and new phenomena may emerge in these broader geometric contexts.

To mitigate these risks, I will adopt an incremental and flexible approach, ensuring that each stage yields publishable results even if full generalization is not immediately possible. If technical challenges arise in the general case, I will focus on special subclasses or lower-order expansions, documenting partial results for future development. Regular collaboration and consultation with experts in CR and contact geometry will provide guidance and alternative perspectives. Should standard microlocal or stationary phase methods stall, I am prepared to explore alternative analytic techniques, such as heat kernel or spectral approaches. Periodic review of progress will allow early identification of bottlenecks and timely adjustment of research direction. Through this adaptive strategy, I am confident that the project will deliver substantial advances in kernel asymptotics and quantization, while laying the groundwork for future exploration in contact and conformal geometry.

The following presents the list and timeline of research milestones (a Gantt chart is also provided for visual reference).

PhD Research Milestones:

Year 1

First Half:

A1. Derive and verify the explicit formulas for the parametrix of the Kohn Laplacian, the Szegő kernel, and the Heat kernel in the Heisenberg model.

Second Half:

A2. Extend the analysis to abstract CR manifolds of codimension one and the CR line bundle case.

B1. Establish a Szegő limit theorem in conformal geometry.

Year 2

First Half:

A2a. Compute the coefficients of the Szegő kernel and related Toeplitz operators.

Second Half:

B2. Construct and analyze Szegő- and Toeplitz-type kernels in contact geometry.



Year 3

First Half:

A3. Generalize the explicit formula to higher-codimension CR manifolds (both model and abstract cases).

Second Half:

B3. Investigate Donaldson–Sun type problems concerning the stability of the Szegő kernel in CR geometry.

Year 4

First Half:

Draft the PhD dissertation.

Second Half:

Revise and defend the PhD dissertation.

Please describe any specialist knowledge or data required to undertake your proposed research, such as language competency, technical skills or use of specialist software. If this knowledge or data is not already in place, details should be provided as to how it will be acquired over the course of the scholarship: **[Max 500 words]**

This proposed research requires advanced specialist knowledge in several interconnected areas. First and foremost, a deep understanding of Fourier integral operator theory is essential, as it forms the analytic backbone for microlocal analysis and the study of kernel asymptotics, particularly in the context of Bergman and Szegő kernels. This theory is crucial for constructing parametrices, analyzing spectral properties, and understanding quantization phenomena in complex, CR, contact, and conformal geometries. While I have a solid foundation in microlocal analysis and pseudodifferential operator theory from my previous studies, I plan to further strengthen my expertise in Fourier integral operators by engaging with classic texts, participating in advanced seminars, and collaborating with experts in the field.

Equally important is proficiency in contact and conformal geometry, which are central to extending analytic techniques beyond the traditional complex and CR settings. This involves mastering the structure of contact manifolds, sub-Riemannian Laplacians, and conformal invariants, as well as learning how symbolic calculus and heat kernel methods adapt to these geometric frameworks. I intend to acquire this knowledge through targeted coursework, reading groups, and close mentorship from faculty members specializing in these areas. The ability to generalize analytic constructions, such as the parametrix of the Kohn Laplacian, to contact and conformal settings will be a key technical challenge and a major focus of my training.

In addition, my research will require technical skills in mathematical software for symbolic and numerical computations. Familiarity with tools such as Python and SageMath will be useful for explicit calculations of kernel coefficients, spectral invariants, and operator



and I will further develop my computational skills as needed, making use of workshops and online resources.

If my research delves into complex geometry questions closely related to differential geometry, such as those arising in Donaldson-Sun theory, I will systematically acquire the necessary background in global analysis, curvature invariants, and deformation theory through advanced courses and collaborative research. This will enable me to bridge microlocal analytic techniques with broader geometric settings and to explore the interplay between symbolic calculus and geometric invariants.

Most research activities will be conducted in English, thus I will improve my language technique, including writing and presentation. Any further specialist knowledge or data required for the project will be acquired through graduate-level courses, departmental seminars, reading groups, and close collaboration with supervisors and peers, leveraging the rich resources and expertise available at the institution. This comprehensive approach will ensure that I am fully equipped to undertake the proposed research and to contribute meaningfully to the advancement of analysis, geometry, and quantization.

Please outline your plans for the dissemination and knowledge exchange of your research, including publications, conference attendance, poster presentations, reports and outreach activities. Details should also be provided as to how the impact of your research will be measured: **[Max 500 words]**

At this stage, my research is still in its early stages, and I have not finalized any specific plans for publication or presentation. However, I am committed to sharing my findings through various channels as my work progresses. I intend to submit my results to leading journals in the fields of complex geometry and microlocal analysis.

To facilitate knowledge exchange, I plan to present at international conferences and workshops that focus on complex analysis, CR geometry, and quantization. Additionally, I will participate in local seminars and collaborative meetings to engage with experts and gain insights from diverse perspectives.

Beyond academic circles, I aim to make my work accessible to a wider audience. For instance, I plan to participate in graduate seminars that introduce geometric analysis and quantization and exchange ideas with early-career mathematicians.

The impact of my research will be evaluated not only through publications and citations but also by the extent of collaboration it fosters and the influence of my methods on related areas. By adopting an open and collaborative approach, I intend to ensure that the outcomes of my research contribute to the broader mathematical community.

Please outline **your** reasons for choosing your proposed (a) primary supervisor(s) and (b) Irish Research Body making particular reference to how the chosen primary supervisor and Irish Research Body:



- can support the acquisition of methods and knowledge necessary to completing the proposed project
- can provide any facilities and infrastructures or other supports relevant to completing the proposed project
- can provide an appropriate and comprehensive training programme

[Max 400 words]

Professor Nikhil Savale is an excellent choice as my primary supervisor because his research expertise closely aligns with the analytic and geometric aspects of my proposed project, particularly in Bergman and Szegő kernels, contact and sub-Riemannian geometry, and quantization. Professor Savale's recent work on kernel asymptotics and geometric quantization provides both the technical depth and innovative perspective necessary for advancing my research, especially as I seek to extend microlocal and Fourier integral operator methods to contact and conformal settings. He could provide a collaborative environment where I can acquire advanced analytic techniques, deepen my understanding of geometric invariants, and receive mentorship in both foundational and cutting-edge aspects of the field.

The Irish Research Body, and specifically the School of Mathematics at Trinity College Dublin, offers an ideal setting for my research. The department hosts leading experts in complex and CR geometry, including Professors Dmitri Zaitsev and Andreea Nicoara, whose work on the Kohn Laplacian and CR analysis provides opportunities to learn different techniques. The department's active research community and international collaborations further provide a stimulating intellectual environment and opportunities for professional development.

In summary, the combination of Professor Savale's mentorship, the expertise of the wider faculty, and the resources and training programmes offered by the TCU will fully support my acquisition of the methods and knowledge necessary to complete my proposed project. This environment will enable me to develop the technical skills, analytic frameworks, and collaborative networks required for success in advanced research at the intersection of analysis, geometry, and quantization.

Please provide details of any proposed research trip(s) of more than four weeks' duration which you believe will be necessary for the successful completion of your scholarship: **[Max 300 words]**

I currently do not have any fixed plans for research trips longer than four weeks. However, as my project progresses, I expect collaborative visits to become crucial. If the need arises, I plan to visit institutions with active groups in CR geometry, Fourier integral operators, and geometric quantization. These visits would enable me to discuss technical challenges directly with experts, learn new analytical techniques, and strengthen collaborations that could extend beyond the scholarship period. While no specific arrangements have been made yet, I will actively seek such opportunities through academic networks and conferences. Any research



trip I undertake will be carefully aligned with the project's objectives to ensure that it significantly contributes to the successful completion of my doctoral research.

The following question applies ONLY to applicants to the Andrew Grene Postgraduate Scholarship in Conflict Resolution, Environmental Protection Agency Postgraduate Scholarship, Postgraduate Scholarship in Childhood Disadvantage in Ireland or the Met Éireann Postgraduate Scholarship. Please outline the reasons for choosing your specific theme, including how your proposal explicitly aligns with the specific requirements as outlined in the theme information: [Max 500 words]

If you have supplementary information, e.g. GANTT chart, diagrams or a bibliography, which accompany your research proposal, please upload them here in PDF format. Supplementary material should only include essential information required for the interpretation and understanding of the proposed research. **Applications will be deemed ineligible and will not be considered for funding if they include additional information about the research proposal or applicant (ie. CV).** Please do not disclose your gender or identity in uploaded PDFs, e.g. use initials instead of first names if you have a publication listed in the bibliography.

Ethics:

Does your research proposal involve any of the following ethical issues of special relevance? 'Informed Consent' or
'Human embryonic stem cells' or
'Privacy and data protection' or
'Use of human biological samples and data' or
'Research on animals' or
'Research in developing countries' or
'Dual use (possible military/terrorist application)' or
'None of the above'

Does your research proposal require approval by the relevant institutional Ethics Committee? 'yes' or 'no'

Please provide a statement detailing the careful consideration you have given to the ethical implications of the proposed research (where ethical issues may arise) and how you plan to address these over the course of your scholarship:

[Max 500 words]

As my research is purely theoretical (mathematics), it does not involve human participants, animals, or sensitive data, and thus, no ethical concerns are expected.

**Sex/gender dimension:**

Please read carefully the section on 'Biological Sex/Gender Dimension' in the Call Document for help in completing this section.

Does your research involve any of the following?

'Humans as the research focus' or

'Animals as the research focus' or

'Human samples and/or data' or

'Humans involved as consumers, users, patients or in trials' or

'Research on animals, animal samples and/or data' or

'Research outputs with implications for both men and women' or

'Research outputs with implications for end users or consumers'

Please provide a statement detailing whether there is a potential sex/gender dimension to be considered in carrying out your research. If your research involves any of the above, please indicate how potential sex/gender issues will be handled. In particular, you are asked to reference the points mentioned in the 'checklist for sex/gender in research content' in the Guide for Applicants:

[Max 500 words]

As my research is purely theoretical (mathematics), it does not involve human participants, and thus, no sex/gender dimension is expected.

Data management

Please provide a data management plan which addresses the following:

- How will data be exploited and/or shared/made accessible for verification and reuse?
- How will data be curated and preserved?
- If applicable, how do you plan to make data FAIR (findable, accessible, interoperable and reusable)?
- If data cannot be made available, why?

[Max 300 words]

My research is purely theoretical and does not involve any personal or sensitive data. Therefore, no data confidentiality or ethical risks are expected. All digital files (e.g., LaTeX, codes, and plots) will be stored on cloud storage with regular backups. Final results will be disseminated through peer-reviewed publications and open repositories when appropriate.



APPLICANT DECLARATION

I hereby declare that I have read and accept the Applicant requirements as set out in the Call Document and General Terms and conditions and Guide for Applicants on the Research Ireland website:

[I agree](#)

I confirm that the information supplied in this application is correct and recognise that should it become apparent that any of the information provided in the application is inaccurate or not verifiable with appropriate documentation, it will result in the application automatically being deemed ineligible:

[I agree](#)

I hereby declare that this application is entirely my own work and understand that it will be subject to plagiarism checks.

Please note that random sampling for evidence of plagiarism will be carried out during the assessment process. Applicants who have been found to plagiarise will be prohibited from applying for Research Ireland funding in the future. Suspected instances of plagiarism will be brought to the attention of the higher education institution:

[I agree](#)

In order to submit your application, click 'Save draft' and then 'Submit Application'.

NOTE: Once the application is submitted, NO further changes can be made.

Semi-Classical and Microlocal Analysis of Szegő Kernels and Toeplitz Quantizations

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PhD Timeline

